

FUP - Oral Exam

neslusam and question providers

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⚠ TRIGGER WARNING - M*NAD ⚠ - The text below contains repeated, explicit occurrences of the “M-Word”. If you are currently recovering from trying to understand what a m*nad actually is, further review of this material is not recommended..

Introduction

[V] - Tomáš Votroubek , [H] - Rostislav Horčík , [Z] - Matěj Zorek **Your examiner is random, your previous lecturer doesn’t make a difference.**

There might be a newer version of this document, check out this [github repository](#)

Functional Programming

[Z] *What is pure/Higher-order function + examples of each possible combination?*

Solution:

Pure function is like a mathematical function. Pure function is a **deterministic function without any side effects** (e.g., no modifying global variables, no writing to files)

Higher-order function: function taking **other functions as arguments** or **returning a function** or both is called a higher-order function.

order \ purity	Pure	Impure
First-Order	circle_area r = pi * r * r	print x / getLine
Higher-order	map / filter / foldl / apply	mapM / modify (State monad)

[Z] *List types of recursions based on branching factor and provide examples. Classify head/tail recursion and discuss theirs branching factor.*

Solution:

Recursion is split into **linear/tree recursion**.

Linear recursion makes **one** recursive call per execution step. Example: **Factorial**

Tree (Non-linear) recursion makes **two or more** recursive calls per execution step.

Example: **Fibonacci**

In **head recursion**, the recursive call is made **before** the rest of the function's operations are completed. Head recursion is **usually non-linear**, because linear recursion is faster with tail recursion.

Example:

```
factorialHead 0 = 1
factorialHead n = n * factorialHead (n - 1)
```

In **tail recursion** the recursive call is the **very last** operation executed by the function. It often uses an **accumulator**. Tail recursion is always **linear**.

Example:

```
factorialTail 0 acc = acc
factorialTail n acc = factorialTail (n - 1) (n * acc)
```

Racket

[V] *How to implement infinite stream of nats/ones*

1. *using stream functions?*
2. *without using stream functions?*

Solution:

1. Example - stream of natural numbers using stream-cons

```
(define (stream_nats n)
  (stream-cons n (stream_nats (+ n 1))))
```

- 2.

```
(define (nats_lambda n)
  (cons n (lambda () (nats_lambda (+ n 1)))))

;;for extra points use syntactic sugar thunk ;)
```

```
(define (nats_thunk n)
  (cons n (thunk (nats_thunk (+ n 1)))))
```

[V] *Describe how does foldl/foldr work, which parametres are used and what is the difference between foldl and foldr.*

Solution:

Foldl/r collapses a list into a single value, combining its elements with an accumulator from left to right (foldl) / from right to left (foldr).

Example:

```
(foldl append '() '((1 2) (3 4) (5 6)))  
;; Output: '(5 6 3 4 1 2)
```

```
(foldr append '() '((1 2) (3 4) (5 6)))  
;; Output: '(1 2 3 4 5 6)
```

λ -Calculus

[Z] List and describe terms in syntax of λ -calculus.

Solution:

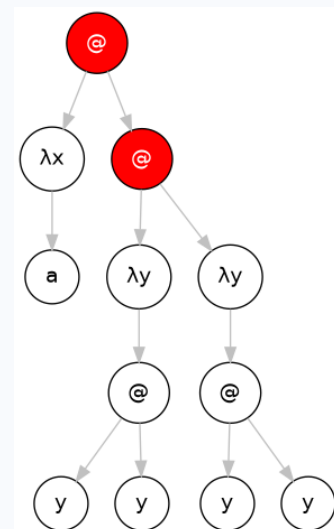
The syntax of lambda calculus has only three types of terms: a **variable** (denoted by lowercase letters $x, y, z, \dots$), the **abstraction** of a variable from a term defining a function (denoted as $\lambda x. t$), and the **application** of a term to a term (e.g. xy , $(\lambda x. y)y$, $\dots$).

[H/Z] What is a *redex*?

Solution:

A **redex** (reducible expression) is any term that **matches abstraction to an argument**. It is similar to having a function (abstraction) that already got the argument and can be evaluated. It is really easy to find in an abstract syntax tree. Redex is **every application**, where the left argument is an abstraction (lambda symbol).

Example: This AST has all the redexes marked red.



[H] What is a reduction? List and describe different types of reductions and their usage.

Solution:

Reduction is the step-by-step process of **evaluating** or simplifying an expression **until it cannot be simplified any further**.

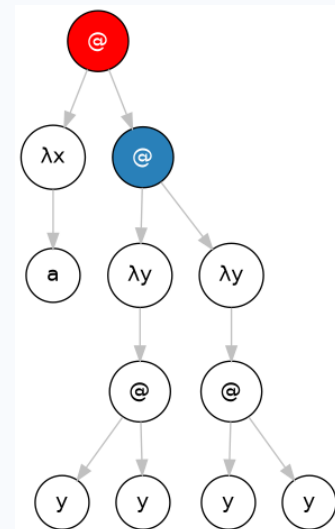
The two essential transformation rules in lambda calculus are:

1. **α -conversion** - The process of **renaming a bound variable** (and all its occurrences inside the function body). Used to **prevent naming conflicts**
2. **β -reduction** - The actual computation step. It applies a function to an argument by **substituting the argument for all free occurrences of the bound variable inside the function body**.

There are **two** different strategies to evaluation:

1. **Left outermost** (Lazy evaluation/normal order) - furthest to the left and closest to the outside of the expression (**not contained inside any other redex**).
2. **Left innermost** (Eager evaluation/applicative order) - furthest to the left but nested the deepest inside the expression (**it contains no other redexes inside itself**).

Example: **Red** - left outermost , **blue** - left innermost

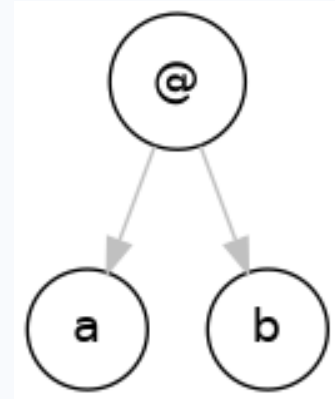
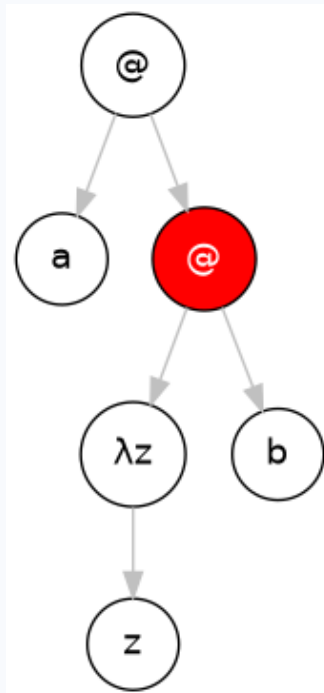
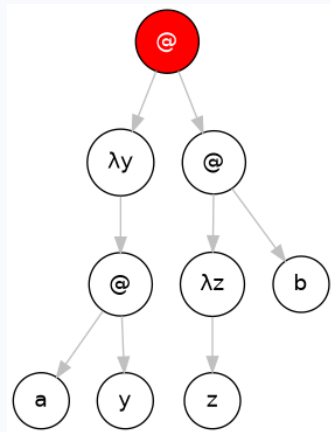


[V] What is a normal form and how to transform a λ -expression to its normal form?

Solution:

Expression is in a **normal form** if it cannot be simplified any further – it contains no redexes. You can transform an expression into its normal (if it exists) form using **repeated β -reductions** until there are no redexes. The preferred evaluation order is **left outermost** (see Church-Rosser theorems to learn why).

Example: transformation of $(\lambda y. ay)((\lambda z. z)b)$



Useful tips:

Be aware of name collision! Perform α -conversion before reducing to keep variable names unique.

Some expression do not converge and do not have a normal form! Example: $(\lambda x.xx)(\lambda x.xx)$

[Z] What is the difference between bound and free variable? Which operation is done on them?

Solution:

Bound variable is inside of an abstraction, free is not.

Possible operations(probably what the teacher meant):

Bound variable - α -conversion - you can rename lambda and each instance in the body

Free variable - β -reduction - argument of a redex destroys the bound variables, only keeping the free ones.

[H] What are the Church-Rosser theorems?

Solution:

In the answers above, we have talked about different evaluation orders/strategies. That gives rise to two natural questions. For a given term, do all evaluation orders produce the same normal form? Is the process of repeated β -reduction guaranteed to terminate for all evaluation orders if the normal form exists? The Church-Rosser theorems give us the answer.

Explicitly, they state that

1. Normal forms are independent of the evaluation strategy – a lambda term cannot be reduced to two different normal forms.
2. Normal order (lazy) evaluation always finds a normal form if it exists.

Informally, if we are to reduce a lambda term, normal order (lazy) evaluation is the way to go. It is guaranteed to produce a result (if the normal form exists) and the result is the same no matter the evaluation order.

[H] *how to represent True/False in λ -Calculus?*

Solution:

T = $\lambda x \lambda y. x$

F = $\lambda x \lambda y. y$

Boolean is interpreted as a **decision between 2 options** and you either pick the first one (T) or the second one (F)

[H] *How to represent number in λ -Calculus? How to represent addition of 2 numbers?*

Solution:

Number n is represented as applying function f to starting x n-times.

$0 = \lambda f. \lambda x. x$ $1 = \lambda f. \lambda x. f(x)$ $2 = \lambda f. \lambda x. f(f(x))$

addition is defined as:

ADD = $\lambda m. \lambda n. \lambda f. \lambda x. m \ f \ (n \ f \ (x))$

Example: $\text{ADD } 1 \ 2 = \lambda f. \lambda x. 1 \ f \ (2 \ f \ (x))$ - **Note! We already defined what 1 and 2 means.**

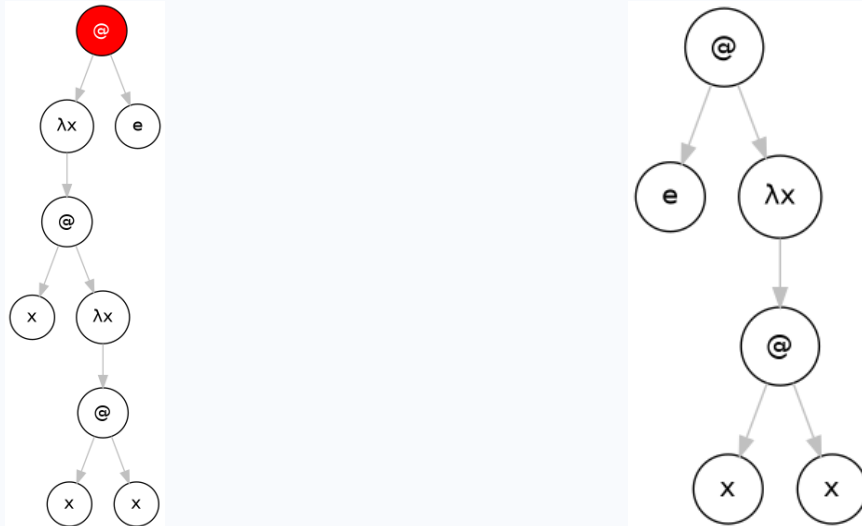
Argument of 1 f is $(2 \ f \ (x)) = f \ (f \ (x))$

We are left with $\lambda f. \lambda x. 1 \ f \ (f \ (f \ (x)))$, which is transformed back to $\lambda f. \lambda x. f \ (f \ (f \ (x))) = 3$

[H] *reduce this expression $(\lambda x.x(\lambda x.xx))e$ to its normal form.*

Solution:

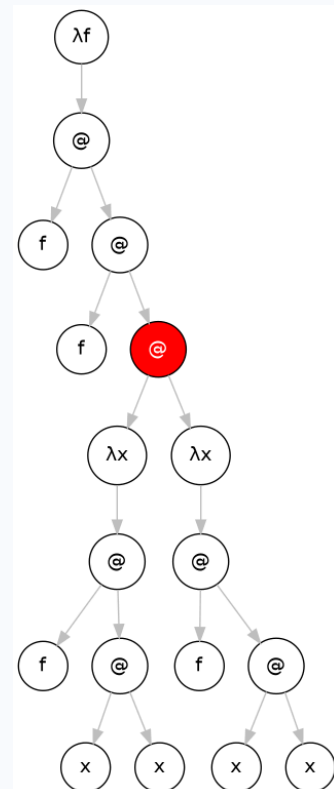
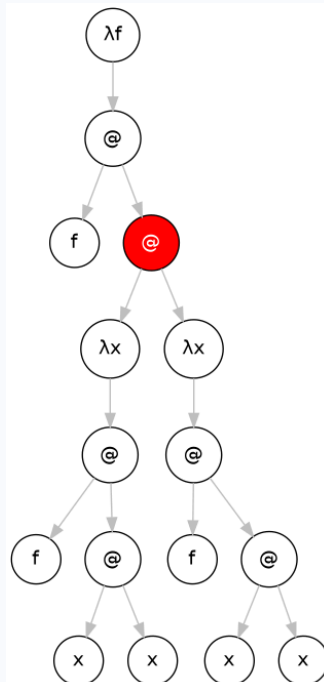
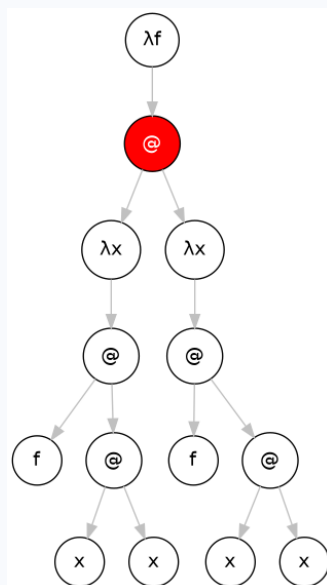
This reduction because is tricky, because there are **two abstractions with the same name**, i would generally recommend to rename one to prevent mistakes from happening.



[V] Find all free/bound vars , redexes in $(\lambda f. (\lambda x. f (x x)) (\lambda x. f (x x)))$. Find normal form.

Solution:

All variables are bound and there is only one redex (marked red). After a few steps, it is obvious that the term **grows infinitely and thus doesn't have a normal form**.



Haskell

[V] Describe signature of function map.

Solution:

```
map :: (a -> b) -> [a] -> [b]
```

[V] Describe signature of function foldl/r.

Solution:

```
foldr :: Foldable t => (a -> b -> b) -> b -> t a -> b
foldl :: Foldable t => (b -> a -> b) -> b -> t a -> b
```

[H] What is a typeclass?

Solution:

Typeclass is a sort of interface that defines some behavior. Typeclasses enable functions to work on different types while supporting certain operations.

For a custom data type, you can usually derive some typeclasses: `Eq`, `Ord`, `Enum`, `Bounded`, `Show`, `Read`. If this is not the case, you need to implement it yourself ;).

[H] How to implement polymorphic typeclass, which adds 1 to an `Int` and prints out length for `String`?

Solution:

You need to create a polymorphic typeclass with a single method that **returns IO monad**.

```
class Magnesium a where
    wheatBran :: a -> IO ()

instance Magnesium Int where
    wheatBran n = do
        let next = n + 1
        print next

instance Magnesium String where
    wheatBran str = do
        let len = length str
        print len
```

[V] What is `Functor` typeclass? Which functions define it?

Solution:

Functor is any data type which can be mapped over. It enables applying pure functions without changing the structure.

Core function: `fmap :: (a -> b) -> f a -> f b` (or the `<$>` operator)

Example:

```
addOne = fmap (+1)

addOne (Map.fromList [('a',1), ('b',2)])
--Output: Map.fromList [('a',2), ('b',3)]
```

[V] What is a `Applicative` typeclass? Which functions define it?

Solution:

`Applicative` typeclass **extends Functor** typeclass. With a `Functor`, you use `<$>` to map a normal function over a wrapped value. With a **`Applicative`** typeclass, you can **apply a function that is already wrapped to a value that is also wrapped** using `<*>`, called **`apply`** or **`ap`**.

Another function of applicative typeclass is to **wrap in the simplest way possible** into a certain data type, using the function **pure**.

Core functions:

```
pure :: a -> f a
```

```
(<*>) :: f (a -> b) -> f a -> f b
```

Examples:

```
-- fmap fails here, because the function is wrapped
```

```
Just (+10) <$> Just 5 -- Result: Error
```

```
Just (+10) <*> Just 5 -- Result: Just 15
```

```
Nothing <*> Just 5 -- Result: Nothing
```

```
pure 5 :: Maybe Int -- Output: Just 5
```

```
pure 5 :: [Int] -- Output: [5]
```

[V] *What is a Monad? Which functions define it?*

Solution:

A Monad **extends Applicative** typeclass. Monads allow to **chain dependent operations**, meaning the output of **step A directly determines what step B will look like**. In contrast, Applicative executes a series of independent operations that are ignorant to each other until they are combined at the very end.

Core Functions:

1. **bind** (written as `>=>`): It feeds a boxed value to a function that returns a new box, without double-boxing. In the `do` notation, it is written with a arrow `<-`

```
(>=>) :: m a -> (a -> m b) -> m b
```

Example:

```
Just 10 >=> (\x -> Just (x + 1)) --Output: Just 11 (Unpacks 10, adds 1, repacks)
```

```
Nothing >=> (\x -> Just (x + 1)) --Output: Nothing
```

```
Just 10 <*> (\x -> Just (x + 1)) --Output: Error (Applicative is not enough)
```

2. **>> operator**: `>>` chains two monadic actions together by running the first, discarding its value, and then running the second. This seems useless at first, but under the hood, every time you write a line in a `do` block without using `<-`, Haskell translates it to `>>` under the hood.

```
(>>) :: m a -> m b -> m b
```

Example:

```
--This is equivalent:
```

```
main = do
```

```

putStrLn "Hello"
putStrLn "World"

--to this:
main = putStrLn "Hello" >> putStrLn "World"

```

3. **return**: the **same function as pure** in Applicative typeclass

```
return :: a -> m a
```

Example from 1. rewritten with do notation.

```

successExample :: Maybe Int
successExample = do
  x <- Just 10
  return (x + 1)

```

[H] What is a State Monad, what is it good for? How to implement bind for State Monad?

Solution:

State Monads simulate mutable state in pure functional programming. State is just a function that takes an initial state *s*, and returns a tuple containing the result *a* and a new state *s*.

```
State s a = s -> (a, s)
```

Even though a student has only been asked about **bind**, I highly recommend looking into `State.hs` provided in the official page. In my opinion, the question about **bind** opens possibilities to be asked about different functions.

```

newtype State s a = S { runState :: s -> (a, s) }

instance Monad (State s) where
  -- (>=) :: State s a -> (a -> State s b) -> State s b
  stx >= f = S (\s -> let (x, s') = runState stx s
                      in runState (f x) s')

```

You are all done!

